

Software Conceptual Design

Next-Generation Ground-based Solar Observing Network

ngGONG Software Team

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Part I

Design

1 Scope and Document Relationships

> **Draft stub:** This section will define the software system boundaries and responsibilities covered by this document.

Planned content includes:

- the scope of High Level Software and Controls (HLS & C) and the Data Management System (DMS), both individually and as an integrated system;
- the boundaries between HLS & C, DMS, other ngGONG systems, and external systems; and
- the relationships between this document and the applicable Design Requirements Documents (DRDs), operational concept documentation, system architecture, and supporting plans.

This section will identify authoritative source documents rather than repeat their requirements, operational narratives, or detailed processes.

2 Architectural Drivers and Constraints

> **Draft stub:** This section will identify the requirements, assumptions, and constraints that materially influence the conceptual software design.

Planned content includes:

- requirements that drive major architectural choices;
- significant performance, capacity, availability, interoperability, information assurance, and lifecycle constraints;
- external dependencies and constraints imposed by other ngGONG systems; and
- assumptions, open questions, and unresolved allocations that affect the design.

Requirements will be referenced by their authoritative identifiers and will not be duplicated here.

3 Architectural Overview

> **Draft stub:** This section will present the integrated conceptual architecture for HLS & C and DMS.

Planned content includes:

- the major software systems, subsystems, and shared services;
- their principal responsibilities and ownership boundaries;
- the major control, science-data, metadata, status, and engineering-data paths; and
- the relationships among station, observatory, and data-management elements.

This section will include or reference the system-context and integrated architecture views needed to explain the design.

4 Major HLS Software Components

> **Draft stub:** This section will describe the responsibilities, decomposition, and relationships of the major HLS & C components at a conceptual level.

The descriptions will identify major responsibilities, interfaces, control and data paths, dependencies, and unresolved design questions without specifying implementation-level APIs, message definitions, or state machines.

4.1 OCS

TBD: Describe the conceptual responsibilities and decomposition of the Observatory Control System (OCS).

4.2 SCS

TBD: Describe the conceptual responsibilities and decomposition of the Station Control System (SCS).

4.3 TuCS

TBD: Describe the conceptual responsibilities and decomposition of TuCS.

4.4 TeCS

TBD: Describe the conceptual responsibilities and decomposition of TeCS.

4.5 GLI and Other Instruments

TBD: Describe the conceptual responsibilities and software relationships of GLI and other instruments.

4.6 Safety Interlock System

TBD: Describe the conceptual architecture of the Safety Interlock System (SIS), including allocated safety functions, safety authority, independence, major components, protected equipment interfaces, and fault-response concepts.

This subsection will reference the SIS DRD, hazard analysis, and cause-and-effect matrix rather than duplicate them.

5 Major DMS Software Components and Data Lifecycle

> **Draft stub:** This section will describe the responsibilities and conceptual decomposition of DMS and the end-to-end lifecycle of ngGONG data.

Planned content includes:

- data ingest and validation;
- processing allocated to DMS;
- metadata and provenance management;
- storage and preservation;
- data discovery and access; and
- principal exchanges with HLS & C and external consumers.

This section will identify ownership boundaries, major data states, and unresolved allocations without defining detailed schemas or database designs.

6 Selected Sequence Diagrams and Conceptual Behavior

> **Draft stub:** This section will show how major software components cooperate in representative operational and failure scenarios.

Candidate scenarios include:

- an observing sequence through data ingest and availability;
- fault handling and recovery; and
- disconnected station operation and reconciliation.

The selected views will explain software cooperation and significant conceptual state behavior. They will reference, rather than repeat, the operational narratives in the applicable operational concept documentation.

7 Deployment and Communications Concepts

> **Draft stub:** This section will describe the conceptual deployment and communications architecture.

Planned content includes:

- placement of major software components at stations and observatory or data-management facilities;
- infrastructure and platform dependencies;
- major network paths and communications patterns;
- trust, availability, and failure boundaries; and
- operation during degraded or disconnected conditions.

This section will not define deployment scripts, detailed network configurations, or implementation-level protocol specifications.

8 Major Internal and External Interfaces

> **Draft stub:** This section will identify the significant interfaces within the software architecture and between ngGONG software and external systems.

For each major interface, planned content includes:

- ownership and responsibility boundaries;
- principal commands, data, metadata, status, or engineering-information exchanges;
- conceptual communication semantics and interoperability constraints; and
- authoritative interface definitions or Interface Control Documents (ICDs), where available.

Particular attention will be given to interfaces between HLS & C and DMS. This section will not include complete ICDs, detailed APIs, Data Distribution Service (DDS) topic definitions, or Protocol Buffer definitions.

9 Information Assurance and Resilience Concepts

> **Draft stub:** This section will summarize the information assurance and resilience implications of the conceptual architecture.

Planned content includes:

- trust boundaries and access concepts;
- protection of software, data, metadata, and control paths;
- availability, fault containment, backup, and recovery concepts;
- resilience during network or service disruptions; and
- applicable constraints, dependencies, assumptions, and open questions.

Detailed cybersecurity configurations, procedures, and implementation controls will remain in their applicable supporting artifacts.

10 Major Design Decisions and Feasibility Evidence

> **Draft stub:** This section will summarize the major architectural decisions and the evidence supporting the technical feasibility of the conceptual design.

Planned content includes:

- alternatives considered, selection criteria, and decision rationale;
- architectural consequences and unresolved issues;
- performance and capacity estimates, including applicable margins;
- technology maturity; and
- relevant analyses, simulations, prototypes, or other supporting evidence.

Detailed trade studies and any separate Architecture Decision Summary will be referenced rather than duplicated. Demonstrated capabilities will be clearly distinguished from estimates, assumptions, and planned work.

11 Technical Risks and Planned Mitigations

> **Draft stub:** This section will identify the principal technical risks that affect the conceptual software design and the work needed to reduce them.

Planned content includes:

- the architectural area or uncertainty associated with each risk;
- the potential design consequence;
- planned mitigation or risk-reduction activity;
- required analysis, prototype, or design evidence; and
- references to the authoritative project risk register.

This section will summarize design implications and will not replace the project risk-management process or controlled risk records.

12 Development Approach and Path to PDR

> **Draft stub:** This section will identify the design work needed to advance the software architecture from the Conceptual Design Review (CoDR) to the Preliminary Design Review (PDR).

Planned content includes:

- conceptual integration and verification approaches;
- major technical dependencies and unresolved design work;
- planned analyses, prototypes, and other risk-reduction activities;
- required requirements allocation and design-compliance evidence; and
- technical inputs needed by cost, acquisition, operations, lifecycle, and disposition planning.

Detailed development processes, verification procedures, schedules, cost estimates, acquisition plans, and operations or disposition plans will remain in their respective documents.

The final placement of Requirements Allocation and Design Compliance content is TBD. Until clarified, this section will reference the authoritative requirements hierarchy, allocation, compliance, and non-compliance records rather than duplicate them.

Part II

Requirements Allocation

13 OCS Requirements Allocation

14 SCS Requirements Allocation

15 DMS Requirements Allocation

16 SIS Requirements Allocation

Part III

Architecture Design Summary